

# TARUN WARRIER

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## EDUCATION

### BITS Pilani, K.K. Birla Goa Campus

B.E. (Hons.) Electronics & Communication Engineering

Oct 2022 – May 2026

Goa, India

## SKILLS

|                            |                                                                                                                                          |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Frameworks</b>          | PyTorch, TensorFlow, OpenCV, scikit-image, Scikit-learn, LightGBM, LangChain, HuggingFace Transformers, NumPy, Pandas, RDKit, Matplotlib |
| <b>Computer Vision</b>     | CNN, Transformers (ViT), image segmentation, classification, feature extraction, data annotation, weak supervision                       |
| <b>3D / Volumetric</b>     | 3D CNNs, volumetric image handling, ROI selection, MR/MRI and CT scan processing                                                         |
| <b>Multimodal</b>          | Multimodal fusion (audio + ECG + imaging), modalities: MR/MRI, CT, ECG, PCG                                                              |
| <b>Deep Learning</b>       | LSTM, MFCC signal processing, local-feature Transformer matching, RGCN                                                                   |
| <b>ML Methods</b>          | RAG, Stacking Ensembles, BERT/CLIP Embeddings, Reinforcement Learning (Double Q-Learning), KNN Imputation, molecular descriptors         |
| <b>AI Applications</b>     | NLP, Edge AI, Medical Diagnostics, Pharmacovigilance, Drug Discovery                                                                     |
| <b>Edge &amp; Hardware</b> | ADI MAX78000FTHR microcontroller, EdgeAI deployment                                                                                      |
| <b>Languages</b>           | Python (primary), Java, C/C++, MATLAB, JavaScript                                                                                        |

## WORK EXPERIENCE

### Data Science Intern

Jan 2026 – Present

Novartis Healthcare

Hyderabad, India

- Built an AI assistant for medical reviewers handling drug safety case reports, aimed at removing inconsistency and improving review speed, reducing processing time from 20–30 minutes to under 1 minute (20x reduction).
- Designed a pipeline that calls multiple AI models with live database searches across Embase and openFDA to retrieve drug safety information, supporting human-reviewed automated assessments of clinical case records.
- Reduced incorrect AI outputs by grounding responses in retrieved source documents; improved retrieval quality and managed input size for accurate analysis of complex safety datasets.

### Research Intern

Jun 2025 – Jul 2025

CiSTUP, IISc Bangalore

Bangalore, India

- Built India's largest annotated vehicle dataset for Indian traffic conditions, with over 10,000 labelled images for vehicle re-identification.
- Applied a deep learning model for local image feature matching on Google Street View imagery for transportation research.

### AI/ML Intern

May 2024 – Jul 2024

Southern Regional Load Dispatch Centre

Bangalore, India

- Built an anomaly detection system for power grid monitoring aimed at identifying the type of fault or anomaly present, achieving 94% classification accuracy using a hybrid CNN-LSTM model.
- Improved detection coverage by combining supervised and unsupervised methods, including time-series clustering for voltage pattern recognition.

## PROJECTS & RESEARCH

### NeuroValve – Molbio Student Innovation Award

Apr 2025 – Present

- Built NeuroValve, a heart disease detection system aimed at early detection of valvular heart disease using Phonocardiograms (PCG) and Echocardiograms together, achieving AUC 0.85 on external PCG validation and AUC 0.80 on external ECG validation.
- Expanded from a single-modality prototype built during the Anveshan Fellowship (deployed on ADI MAX78000FTHR microcontroller, 95% internal accuracy on PCG) to a full multimodal system.

### StackTox: Molecular Toxicity Prediction

Manuscript in preparation

- Achieved AUC 0.721 on an independent external test set, outperforming MolToxPred, eToxPred, and ToxinPredictor using only 61 features compared to 944–2,237 in competing models.
- Built StackTox for molecular toxicity prediction, aimed at reducing the time and cost of lab-based screening in drug discovery and chemical safety.
- Trained a stacking model (Random Forest + LightGBM + SVM + logistic regression) on a dataset of 16,192 molecules.

### Aneurysm Detection (3D Volumetric Imaging)

- Built an intracranial aneurysm detection system aimed at identifying the type of aneurysm present using high and low-resolution MR and CT scans of the Circle of Willis, achieving internal AUC of 0.80.
- Used a two-stage approach: region-of-interest selection followed by a 3D CNN for classification.

### AI Knowledge Graph Intelligence

github.com/tarundevs/Multimodal-RGCN

- Built a knowledge graph system using a graph neural network with text and image embeddings for multi-category classification, achieving 79% validation accuracy.

### Reinforcement Learning Game AI

github.com/tarundevs/DRL-TicTacToe

- Built a self-learning game agent using Double Q-Learning that achieved an 88% win rate through trial-and-error training.

## AWARDS & RECOGNITION

### Molbio Student Innovation Award

Sponsored by Molbio Diagnostics for AI research in medical diagnostics

Apr 2025